

Excercise 1

SingletonTest.Java

```
public class SingletonTest {  
    public static void main(String[] args) {  
        Logger logger1 = Logger.getInstance();  
        Logger logger2 = Logger.getInstance();  
        logger1.log("Application Started");  
        logger2.log("Application Running");  
        System.out.println("logger1 hashCode: " + logger1.hashCode());  
        System.out.println("logger2 hashCode: " + logger2.hashCode());  
        if (logger1 == logger2) {  
            System.out.println("Only one Logger instance exists.");  
        } else {  
            System.out.println("Multiple Logger instances exist.");  
        }  
    }  
}
```

Logger.Java

```
public class Logger {  
    private static Logger instance;  
    private Logger() {  
        System.out.println("Logger instance created.");  
    }  
    public static Logger getInstance() {  
        if (instance == null) {  
            instance = new Logger();  
        }  
        return instance;  
    }  
    public void log(String message) {
```

```

        System.out.println("LOG: " + message);
    }
}

```

Execution Image

```

J SingletonTest.java X J Logger.java
J SingletonTest.java > SingletonTest
1 public class SingletonTest {
2
3     Run | Debug
4     public static void main(String[] args) {
5
6         Logger logger1 = Logger.getInstance();
7         Logger logger2 = Logger.getInstance();
8
9         logger1.log(message: "Application Started");
10        logger2.log(message: "Application Running");
11
12        System.out.println("logger1 hashCode: " + logger1.hashCode());
13        System.out.println("logger2 hashCode: " + logger2.hashCode());
14
15        if (logger1 == logger2) {
16            System.out.println(x: "Only one Logger instance exists.");
17        } else {
18            System.out.println(x: "Multiple Logger instances exist.");
19        }
20    }
21}

```

```

PS C:\Users\karun\OneDrive\Desktop\PROGRAMS> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.5.11-hotspot\bin\java.exe' '-XX:+showCodeDetailsInExceptionMessages' '-cp' 'C:\Users\karun\AppData\Roaming\Code\User\workspaceStorage\e08bdf456caae23656b23795694dcde\redhat_java\jdt_ws\PROGRAMS_2cd066f4\bin' 'SingletonTest'
AppData\Roaming\Code\User\workspaceStorage\e08bdf456caae23656b23795694dcde\redhat_java\jdt_ws\PROGRAMS_2cd066f4\bin' 'SingletonTest'
LOG: Application Started
LOG: Application Running
logger1 hashCode: 140435067
logger2 hashCode: 140435067
Only one Logger instance exists.
PS C:\Users\karun\OneDrive\Desktop\PROGRAMS>

```

Excercise 2

Execution Image

```

EXPLORER
COGNIZANT PROGRAMS
J Document.java
J DocumentFactory.java
J ExcelDocument.java
J ExcelDocumentFactory.java
J FactoryMethodTest.java
J PdfDocument.java
J PdfDocumentFactory.java
J WordDocument.java
J WordDocumentFactory.java
J PdfDocumentFactory.java
J FactoryMethodTest.java X J ExcelDocument.java J DocumentFactory.java J WordDocumentFactory.java J PdfDocumentFactory.java
J FactoryMethodTest.java > FactoryMethodTest
1 public class FactoryMethodTest {
2
3     Run | Debug
4     public static void main(String[] args) {
5
6         DocumentFactory wordFactory = new WordDocumentFactory();
7         Document wordDoc = wordFactory.createDocument();
8         wordDoc.open();
9
10        DocumentFactory pdfFactory = new PdfDocumentFactory();
11        Document pdfDoc = pdfFactory.createDocument();
12        pdfDoc.open();
13
14        DocumentFactory excelFactory = new ExcelDocumentFactory();
15        Document excelDoc = excelFactory.createDocument();
16        excelDoc.open();
17    }
18}

```

```

PS C:\Users\karun\OneDrive\Desktop\cognizant programs> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.5.11-hotspot\bin\java.exe' '-XX:+showCodeDetailsInExceptionMessages' '-cp' 'C:\Users\karun\AppData\Roaming\Code\User\workspaceStorage\f0ca74aaa94848f1ce8ce235599e62\redhat_java\jdt_ws\cognizant programs_b4725b91\bin' 'FactoryMethodTest'
FactoryMethodTest
Opening word Document
Opening PDF Document
Opening Excel Document
PS C:\Users\karun\OneDrive\Desktop\cognizant programs>

```